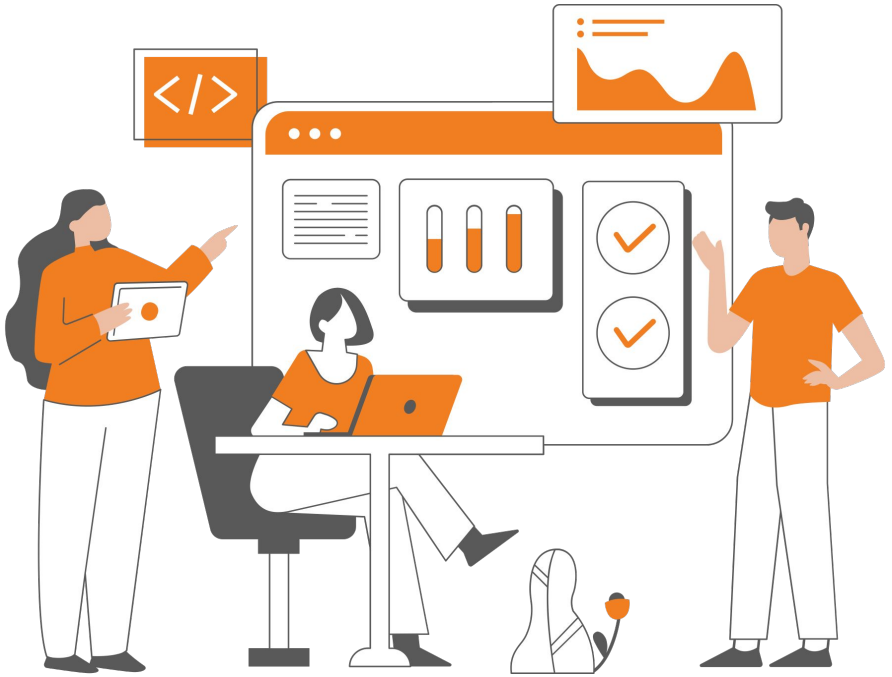
Training On

Introduction of Software Testing:

Software Testing Phase



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What is Software Testing?

- Software testing is the process of checking if a software application as its should and meet the set requirements. This can be done manually by people or with automated tools. The main goal of software testing is to find any errors, missing features, or differences from what was expected.

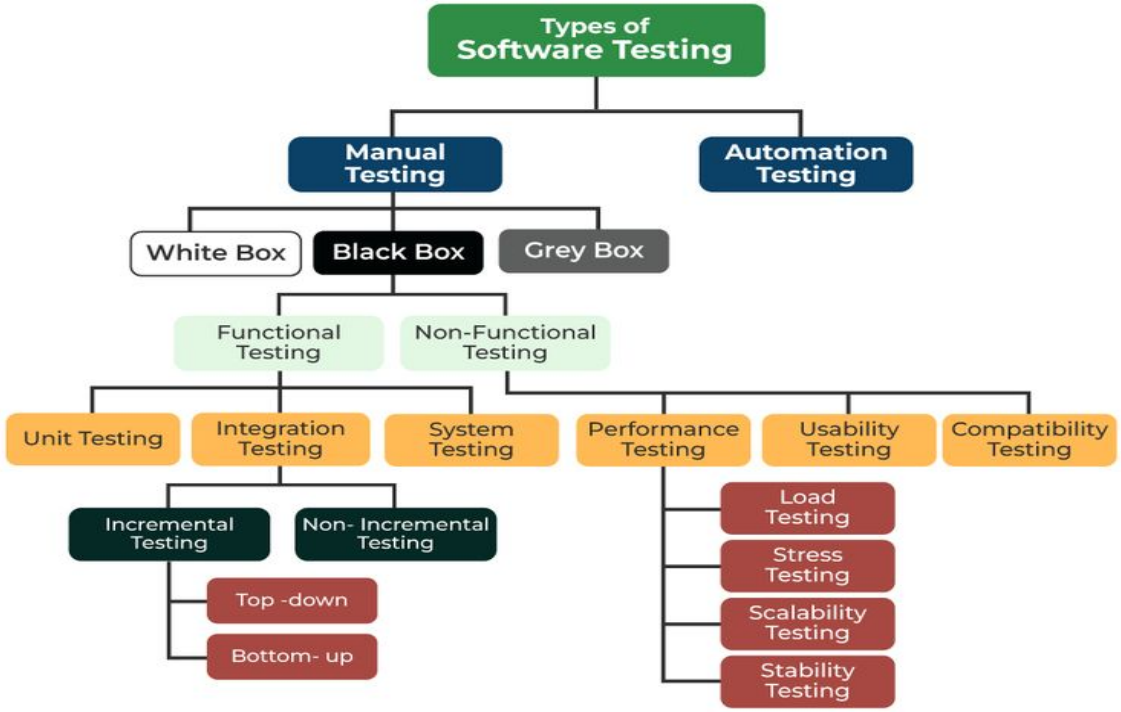
Software Testing

- Types of Testing
- Manual Testing and Automation Testing
- Testing Tools (Postman, JMeter, Selenium Webdriver, etc.)
- Regression Testing
- Web Application Testing
- Desktop Application Testing
- Mobile Application Testing
- Game Testing
- Test Report
- Project Management Tool (Jira, Slack, Trello, etc.)

Different Types of Software Testing

Most common Software Testing techniques

- Black Box Testing
- White Box Testing
- Grey Box Testing
- Functional Testing
- Non-Functional Testing



Types of Software Testing

- **Black Box Testing:**

  It is a type of software testing technique that focuses on testing the functionality of a software system without any knowledge of its internal workings.

- **White Box Testing:**

  White box testing is a method of testing software where the tester knows the internal workings of the application. This type of testing focuses on checking the actual code, logic, and structure to ensure everything works correctly. It involves examining the code to verify paths, conditions, and data flow.

- **Grey Box Testing:**

  Grey box testing is a software test technique that combines the approaches of both black box and white box testing.

Types of Software Testing

- **Functional Testing**
    1. Unit Testing
    2. Integration Testing
    3. System Testing
    4. Acceptance Testing
- **Non-Functional Testing**
    1. Security Testing
    2. Performance Testing
    3. Usability Testing
    4. Compatibility Testing

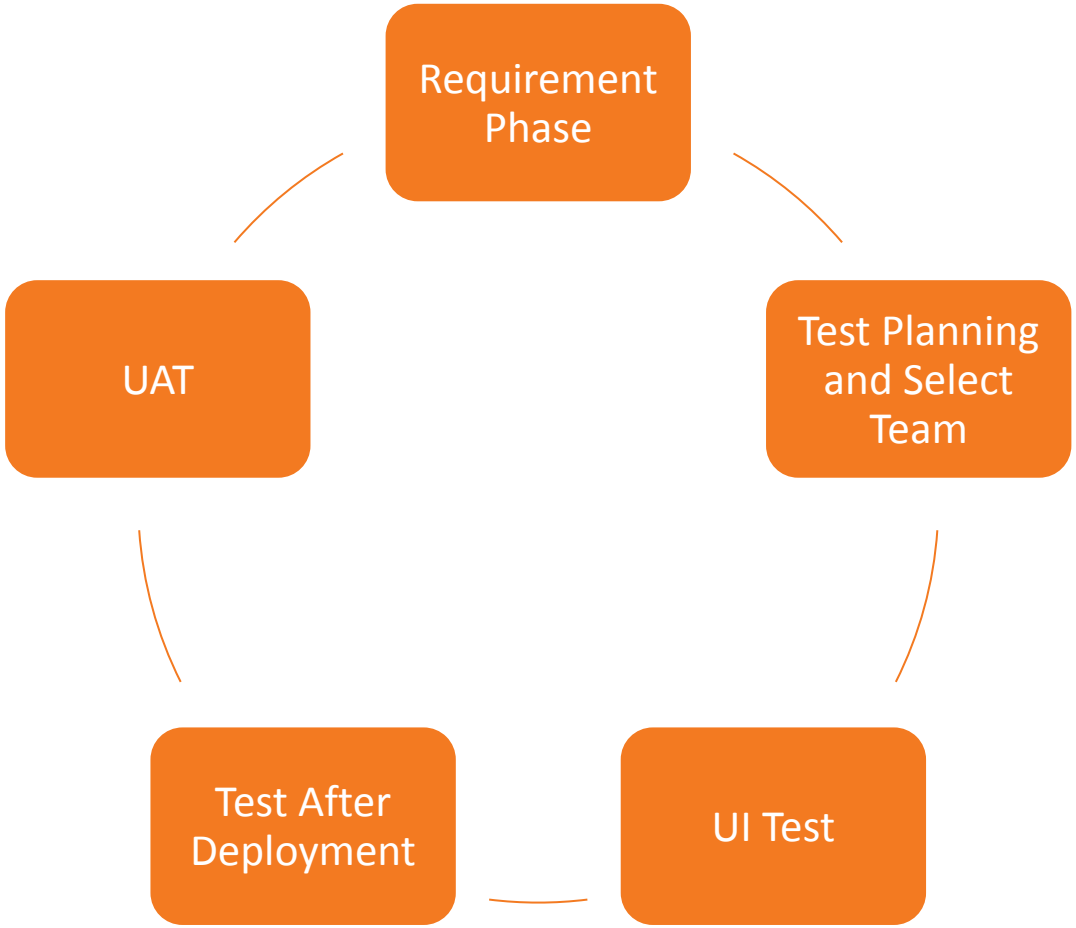
There are many more other types testing like Regression testing, API Testing, Gui Testing etc.

Manual Testing and Automation Testing

- **Manual Testing:** Manual testing involves a human tester manually executing test cases without using any automation tools. The tester checks the software's functionality by following specific steps and comparing the results to the expected outcomes. This type of testing is usually done in the early stages of the software development process. While manual testing can be time-consuming and resource-intensive, it helps identify defects and issues that might be missed by automated tests.
- **Automation Testing:** Automation testing uses tools to run test cases automatically. It is typically performed after manual testing when the software is stable. Testers use scripts to automate repetitive tasks, allowing tests to be completed much faster than manual testing. Automation testing is ideal for large, complex systems and software that requires frequent retesting. Although it saves time and resources in the long run, setting up and maintaining the automation framework can be costly.

QA Process Flow

- Requirement Phase
- Test Plan and Select Team
- UI Test
- Test After Deployment
- UAT
- Example: [QA Brains]



Web Application Testing

- All Functionality
- Responsive(Tab, Mini Tab, Mini Phone, Multiple Browser)
- Regression Test
- Load Test
- Security Testing
- User Activity Log
- Proxy Server
- Social Media Sync (If any)
- Others

Mobile Application Testing

- Splash Screen
- Safe Area
- Screen (Area + others)
- All Functional Area
- Responsive(Tab , mini tab, mini phone, etc)
- Regression Testing
- Load Testing
- Usage (GPU Usage, CPU Usage)
- Device Memory & Battery Optimization
- Test with Low end Device and fps Test, lag Issues
- Security Testing
- User Activity Log
- Proxy Server
- Social Media Sync (If any)
- Others

Game Testing

- Splash Screen
- Safe Area
- Screen (Area + others)
- All Functional Area
- Game Play Test
- Score Submit Functionality
- Leader board/ Share/ Social Share/ Rate Us
- Regression Testing
- Online/ Offline (Mobile Data and Wifi)
- Game Profile(Profile Name, Number or Other things Edit(Validation + Negative + Positive Testing)
- Responsive(Tab , mini tab, mini phone etc)

Game Control System

- Keyboard for Movement: (W, A, S, D), (Upper Arrow, Lower Arrow, Right Arrow, Left Arrow)
- Mouse
- Joy Stick
- Gamepad

AD Operation

- AD (Banner AD position)
- Game over AD
- Game Over Screen
- Game Over Background
- Different Types of AD Testing Like Rewarded Ad, Interstitial Ad [Based On Requirement]
- Ad cooldown time
- Ad appearance after internet on/off

Performance and Security Test

- Load Testing
- Usage (GPU Usage, CPU Usage)
- Device Memory & Battery Optimization
- Test with Low end Device and fps Test, lag Issues
- Security Testing
- User Activity Log
- Proxy Server
- Social Media Sync (If any)

Analytics Check

- Google Analytics
- Firebase Analytics
- Firebase Crashlytics

Test Plan

- A test plan is a document that outlines how, what, and when to test in a software project. It acts as a guide for testing, making sure every part of the application is checked thoroughly. The test plan helps ensure the software meets requirements and is defect-free before release.
- Demo Test Plan

Test Cases

- Test cases are detailed steps to check if a software feature works as expected. Each test case includes what to input, how to test it, and what the expected result should be.
- Demo Test Cases

Challenges During Testing. . .

Prerequisite of Automation Testing

- Download and Install JDK(Java Development Kit)
- Install Eclipse from the official website
- Download the Selenium Java Client version
- Configure the drivers depending on the browser. The example here will be using a chromedriver for Chrome

Before performing automation testing for the login functionality, there are a couple of basic steps that need to be followed for the test case to be written:

- Create a Selenium WebDriver instance
- Configure browser if required
- Navigate to the required web page
- Locate the relevant web element
- Perform action on the web element

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Automation Testing

- Dependencies demo for Selenium

```
<!--
https://mvnrepository.com/artifact/io.github.bonigarcia/webdrivermanager  -->
<dependency>
    <groupId>io.github.bonigarcia</groupId>
    <artifactId>webdrivermanager</artifactId>
    <version>5.6.3</version>
</dependency>
<!--
https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-firefox-driver  -->
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-firefox-driver</artifactId>
    <version>4.16.1</version>
</dependency>
```

Automation Testing

- Login page automation demo

```
2
3       @Test(priority = 1) // LOGIN
4       public void login() throws InterruptedException, Exception {
5
6           System.out.println("WELCOME TO VAT PROMPT LOGIN PAGE");
7           Thread.sleep(300);
8           System.setProperty("webdriver.gecko.driver", "C:\\Users\\user\\Desktop\\geckodriver.exe");
9           driver = new FirefoxDriver();
10          driver.get("http://120.50.6.185:81/LogIn.aspx");
11          WebElement username = driver.findElement(By.id("ContentPlaceHolder1_txtUserId"));
12          WebElement password = driver.findElement(By.id("ContentPlaceHolder1_txtPassword"));
13          WebElement login = driver.findElement(By.xpath("//*[@id=\"ContentPlaceHolder1_btnLogin\"]"));
14          username.sendKeys("test");
15          password.sendKeys("123");
16          WebElement login1 = driver.findElement(By.xpath("//*[@id=\"ContentPlaceHolder1_btnLogin\"]"));
17          login1.click();
18          String actualUrl = "http://120.50.6.185:81/Bss_unt.aspx";
19          String expectedUrl = driver.getCurrentUrl();
20          AssertJUnit.assertEquals(expectedUrl, actualUrl);
21          if (actualUrl.equalsIgnoreCase(expectedUrl)) {
22              System.out.println("LOGIN PASSED");
23          } else {
24              System.out.println("LOGIN FAILED");
25          }
26      }
```

How To Become A Quality Assurance Engineer?

Assessment

1. Write 10 Test cases
2. Write a difference between Sanity testing and Regression testing
3. Annotate Exploratory testing, Happy path Testing, Unit Testing and Functional Testing
4. Run a Login and Registration Automation of QA Brain with Selenium

Last date of submission 12 July, 2024

Assessment Submission Folder



Thank You